

Langerhans Cell Histiocytosis (LCH)

The reactive versus neoplastic nature of Langerhans cell histiocytosis (LCH) remains a subject of debate. However, the most probable hypothesis is that LCH arises from a clonal proliferation of histiocytes in genetically predisposed individuals.

Histopathology

Histopathologic features of LCH include a dense infiltrate of histiocytes with prominent epidermotropism. The hallmark cell is the "**LCH cell**"—a histiocyte characterized by an irregular, vesiculated, often reniform (kidney-shaped) nucleus and abundant, slightly eosinophilic cytoplasm. These cells are immunophenotypically positive for **S100 protein**, **CD1a**, and **CD207 (langerin)**, and contain cytoplasmic **Birbeck granules** (Langerhans granules).

Clinical Features

Cutaneous manifestations of LCH are highly variable and may present as papules, vesicles, pustules, nodules, or ulcers. The disease course ranges from localized, self-healing forms to generalized, life-threatening presentations. Common sites of skin involvement include the **head, trunk, and skinfolds**. Mucosal lesions typically appear as **ulcerated nodules**, predominantly affecting the **gingiva** and **genital regions**.

Systemic Involvement

LCH can involve multiple organ systems. Associated manifestations include: - **Diabetes insipidus** (due to pituitary involvement) - **Exophthalmos** (from orbital bone lesions)

Systemic organs that may be affected include: - Bones - Lungs - Bone marrow - Liver - Spleen - Lymph nodes

Aggressive forms of LCH can be treated with systemic chemotherapy, including **vinblastine** and other agents, with variable success.

Disease Continuum and Classification

It is increasingly realistic to view LCH as a **clinical continuum**, as each patient presents uniquely in terms of lesion type, number, and location. While historical terminology persists for didactic use, the critical distinction lies in identifying: - **Multisystem disease (extensive LCH)**: Requires systemic therapy - **Single-system disease (restricted LCH)**: May be managed with topical treatment or observation - **Self-healing forms**: Often resolve spontaneously

Diagnostic Guidelines

The Writing Group of the Histiocyte Society has established confidence levels for the diagnosis of **Class I histiocytosis (LCH)**, emphasizing the importance of immunohistochemical and ultrastructural confirmation.

Classification of Histiocytosis

TABLE 148-1 Classification System for Histiocytosis (Histiocyte Society, 1987 and 2004)

- **Class I:** Langerhans cell histiocytosis
- **Class II:** Non-Langerhans cell histiocytosis (mononuclear phagocyte system)
 - **IIa:** Histiocytosis involving dermal CD68+ and factor XIIIa+ dendrocytes
 - **IIb:** Histiocytosis involving cells other than Langerhans cells or dermal dendrocytes
- **Class III:** Malignant histiocytosis

This classification provides a framework for diagnosis and management, guiding therapeutic decisions based on disease extent and organ involvement.